

Topic 7.3 Naïve Bayes Classifier

- Based on Bayes' theorem
- Statistical classifier
- Supervised learning
- Performance comparable to decision tree and neural network
- High accuracy and speed when applied to large datasets
- Naïve: Simplified assumption – ‘effect of one attribute is independent of the effect of other attributes’

❖ Basic ideas and major steps

- 1) Each data sample is represented by a feature vector,
 $V = (v_1, v_2, \dots, v_n)$, n - no. of attributes
- 2) Unknown sample, $X = (x_1, x_2, \dots, x_n)$; Classifier predicts that X belongs to one of m classes, C_i with highest posterior probability

$$P(C_i | X) > P(C_j | X), 1 \leq j \leq m \text{ \& } j \neq i. \text{ [Maximum posterior probability]}$$

It is computed using Bayes' theorem as follows:

$$P(C_i | X) = (P(X | C_i) \times P(C_i)) / P(X)$$

- 3) $P(X)$ is constant for all classes, so, $P(X | C_i) \times P(C_i)$ needs to be maximized. If classes are equally likely, $P(C_i)$ can also be dropped, otherwise

$$P(C_i) = S_i / S, \text{ where } S_i - \text{no. of samples of class } C_i, S - \text{total no. of samples.}$$

- 4) Discarding attribute dependence,

$$P(X | C_i) = \prod_{k=1:n} P(x_k | C_i).$$

And $P(x_1 | C_i), P(x_2 | C_i), \dots, P(x_n | C_i)$, easily estimated from training samples.

- For categorical attribute A_k , $P(x_k | C_i) = S_{ik} / S_i$, where S_i - no. of samples of class C_i and S_{ik} - those from S_i with attribute value x_k .
- For continuous A_k , Gaussian distribution is typically assumed:
 $P(x_k | C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i})$ [Gaussian normal density function for A_k , while μ_{C_i} – mean and σ_{C_i} – standard deviation of samples with x_k of class C_i]

Example: We take the same training dataset as for decision tree learning.

ID	Age	Income	Student	Credit Rating	Decision/ Class/ Label
1	≤ 30	high	no	fair	negative
2	≤ 30	high	no	excellent	negative
3	31...40	high	no	fair	positive
4	> 40	medium	no	fair	positive
5	> 40	low	yes	fair	positive
6	> 40	low	yes	excellent	negative
7	31...40	low	yes	excellent	positive
8	≤ 30	medium	no	fair	negative
9	≤ 30	low	yes	fair	positive
10	> 40	medium	yes	fair	positive
11	≤ 30	medium	yes	excellent	positive
12	31...40	medium	no	excellent	positive
13	31...40	high	yes	fair	positive
14	> 40	medium	no	excellent	negative

- We take, C_1 : 'Buys a computer' / 'positive' and C_2 : 'Does not buy a computer' / 'negative'.
- The unknown sample we want to classify is
 $X = (\text{age} = 22, \text{income} = \text{'medium'}, \text{student} = \text{'yes'}, \text{credit_rating} = \text{'fair'})$

- $P(C_1) = 9/14 = 0.643$
 $P(C_2) = 5/14 = 0.357$
- We now compute $P(X | C_i)$, for $i = 1, 2$ as follows:

$P(\text{age} = \leq 30 C_1) =$	$2/9 = 0.222$
$P(\text{age} = \leq 30 C_2) =$	$3/5 = 0.600$
$P(\text{income} = \text{'medium'} C_1) =$	$4/9 = 0.444$
$P(\text{income} = \text{'medium'} C_2) =$	$2/5 = 0.400$
$P(\text{student} = \text{'yes'} C_1) =$	$6/9 = 0.667$
$P(\text{student} = \text{'yes'} C_2) =$	$1/5 = 0.200$
$P(\text{credit_rating} = \text{'fair'} C_1) =$	$6/9 = 0.667$
$P(\text{credit_rating} = \text{'fair'} C_2) =$	$2/5 = 0.400$

- So, we obtain,
 $P(X | C_1) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$
 $P(X | C_2) = 0.600 \times 0.400 \times 0.200 \times 0.400 = 0.019$

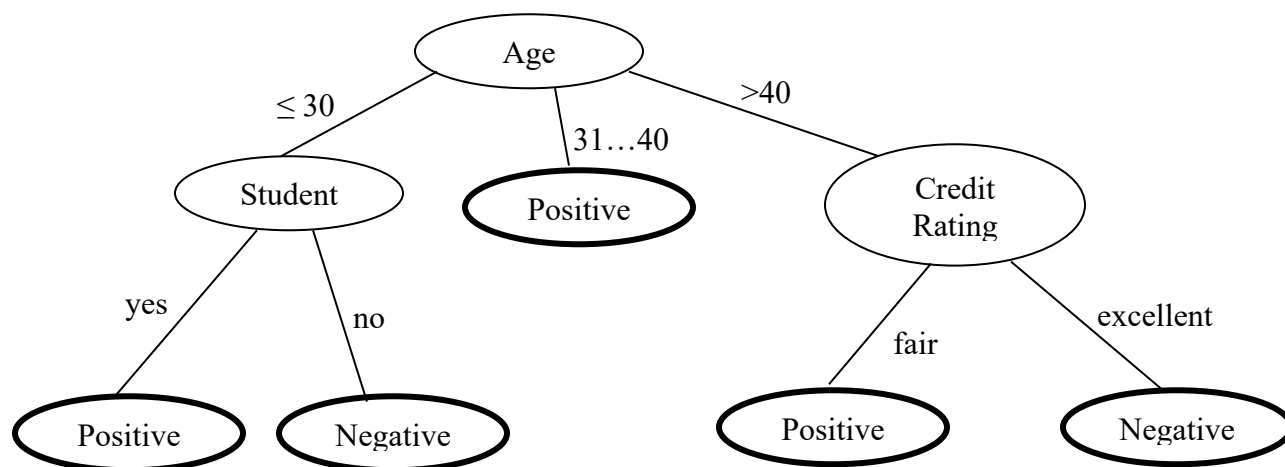
and

$$P(X | C_1) P(C_1) = 0.044 \times 0.643 = \mathbf{0.028}$$

$$P(X | C_2) P(C_2) = 0.019 \times 0.357 = 0.007$$

That is, prediction for sample X is 'positive' ('Buys a computer').

It is the same with decision tree prediction:



If 'Age' = '≤ 30' and 'Student' = 'yes', then 'Class' = 'Buys a computer'.